STEP 1A

SELECTING BRAINSTORMING METHODS

LITERATURE SURVEY

✓ N = 30 structured methods

SELECTED METHODS

FW **Futures Wheel:** divergent futures-based.

EM **Empathy Mapping**: convergent stakeholder-based.

FF **Free-form**: baseline method with minimal structure.

STEP 1B

IDENTIFYING AI POINTS OF INTERVENTIONS (POIs)

LITERATURE SURVEY

✓ N = 23 papers from ACM DL

KEY POI DIMENSIONS

Timing: ideate, converge, or decide
AI role: peer, expert, or facilitator
HAS: AI-driven, equal partnership, or human-driven

STEP 2

DERIVING DESIGN REQUIREMENTS

45 min Ideate | 15 min Converge | 20 min Decide | 80 min

FORMATIVE STUDY (2 WS, NO AI)

✓ N = 12 participants FW EM

CO-DESIGN STUDY (3 WS, WITH AI)

✓ N = 16 participants FW EM FF

FINAL REQUIREMENTS

R1 **Provide hints**
ideate · peer · human-driven

R2 **Generate and refine ideas**
ideate and converge · expert · partnership

R3 **Organise ideas**
converge · facilitator · partnership

R4 **Reference external information**
all phases · expert · AI-driven

STEP 3

EVALUATING AI INTERVENTIONS

45 min Ideate | 15 min Converge | 20 min Decide | 80 min

MAIN STUDY (10 WS)

CHATBOT COMPANION (6 WS)

✓ N = 29 participants FW EM FF

Outcome: AI interventions enhanced output quantity, quality, and workshop experience.

MEDICAL AL (4 WS)

✓ N = 25 participants FW EM

Outcome: AI interventions enhanced output quantity but not workshop experience.